

Running the Project

1. Install Dependencies

```
pip install -r requirements.txt
```

2. Start the Application

Run the main dashboard:

```
python app.py
```

You will see the following menu:

```
=====
RESTAURANT RESOURCE PLANNING DASHBOARD
=====

1. Generate Customer Forecast
2. Generate Staff Schedule
3. Generate Inventory Order
4. View Forecast
5. View Staff Schedule
6. View Inventory
7. Submit Feedback
8. Retrain Model
9. Exit
```

Application Workflow

Follow the steps below in order.

Step 1 – Generate Customer Forecast

Select:

1. Generate Customer Forecast

This predicts the expected customer covers for each operating hour and saves the results to:

```
data/forecast.csv
```

Step 2 – Generate Staff Schedule

Select:

2. Generate Staff Schedule

The application reads the hourly customer forecast and generates a staff schedule by role.

Output:

```
data/staff_schedule.csv
```

Step 3 – Generate Inventory Order

Select:

3. Generate Inventory Order

The application calculates ingredient order quantities using:

- Customer forecast
- Recipe requirements
- Current inventory
- Ingredient shelf life
- Supplier lead time

Output:

```
data/inventory_order.csv
```

Step 4 – View Generated Reports

The dashboard provides options to view generated reports.

Customer Forecast

4. View Forecast

Staff Schedule

5. View Staff Schedule

Inventory Order

6. View Inventory

Step 5 – Submit Manager Feedback

After restaurant operations are complete, select:

7. Submit Feedback

Enter the actual customer covers and the reason for any difference between the prediction and the actual result.

Examples:

- Rain
- Special Event
- Holiday
- Equipment Failure

The feedback is saved to:

data/feedback.csv

Step 6 – Retrain the Forecast Model

Select:

8. Retrain Model

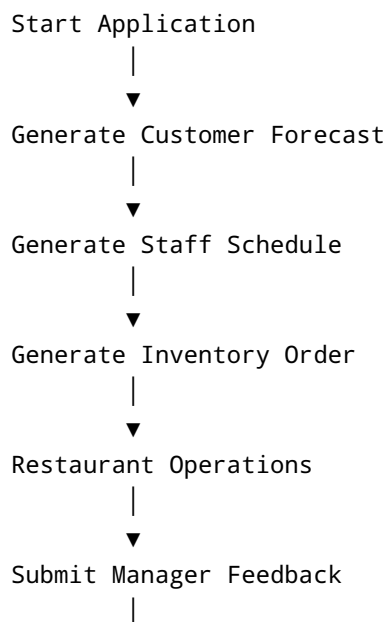
The application retrains the XGBoost forecasting model using historical data and the recorded feedback.

This implements the self-learning feedback loop required by the assignment.

Generated Output Files

File	Description
forecast.csv	Hourly customer cover forecast
staff_schedule.csv	Staff requirements by hour
inventory_order.csv	Ingredient order recommendations
feedback.csv	Manager feedback records

Typical Daily Workflow





Retrain Forecast Model